

SMART INDIA HACKATHON 2026



NexFlow

Problem Statement ID — SIH26205

Problem Statement Title : Student Innovation: address growing pressure on city resources, transport networks & logistics infrastructure

Theme — Transportation & Logistics

PS Category — Software

Team ID — _____

Team Name — NexGen

**PILOT CONTEXT • KOLKATA BURRABAZAR / POSTA
TRADE DISTRICT**

The urban-freight problem is not just a routing problem.

Freight pressure → curb access → live fleet dispatch → measurable outcomes

A truck can take the fastest road and still lose time because the loading bay is occupied. NexFlow optimizes the road + the curb together.

**NEXFLOW • URBAN FREIGHT
INTELLIGENCE LAYER**

PREDICT

ALLOCATE

REROUTE





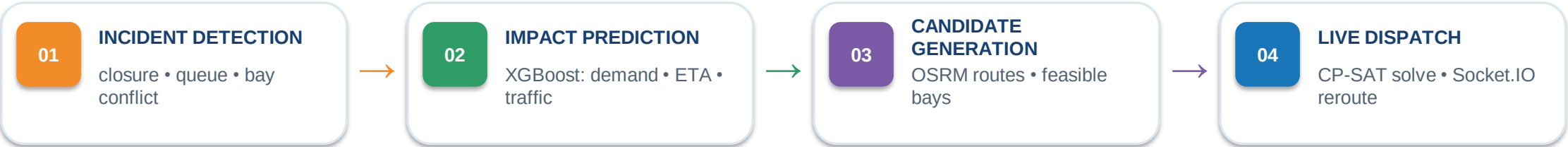
PROPOSED SOLUTION & OPTIMIZATION LOGIC



CORE PITCH • URBAN FREIGHT BOTTLENECKS + LOADING-BAY DYNAMICS

Predict freight pressure → jointly assign vehicle + route + loading-bay slot before curb demand peaks.

FOUR-STEP EXECUTION FLOW



INTUITIVE COST MINIMIZATION

MINIMIZE
Delivery Delay + Curb Waiting + Late Penalties + Unnecessary Distance + Rerouting Cost

SUBJECT TO • vehicle capacity • delivery windows • bay exclusivity • road closures

LIVE DEMO • INTERACTIVE SIMULATION ENGINE

Simulated road closure on a real OSM segment
→ OSRM blocks segment → CP-SAT re-solves fleet + bay slot

Socket.IO animates the vehicle fleet reroute on the map in seconds.

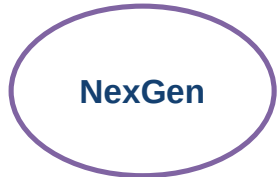
STANDARD NAVIGATION

Point-to-point fast path for one trip

NEXFLOW

Joint route + curb allocation across fleet and bay capacity

MVP solver: OR-Tools CP-SAT rolling-horizon 0–1 assignment model | Full mathematical MILP formulation reserved for Q&A footer on Slide 6.



TECHNICAL APPROACH

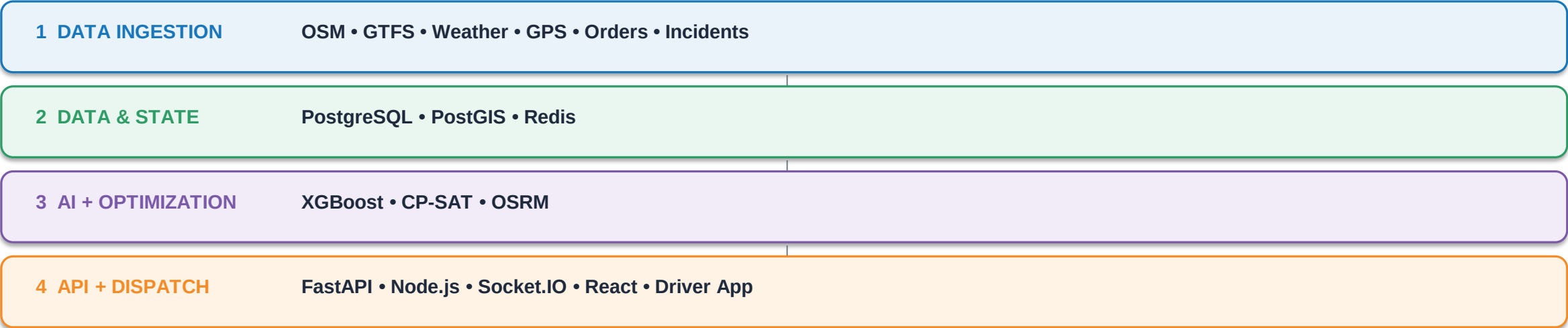


GOVERNMENT-READY ARCHITECTURE

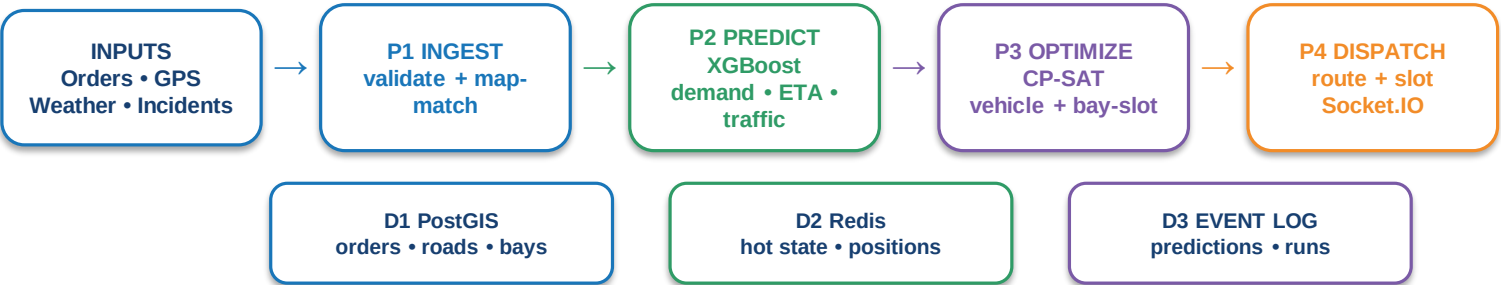
PM GatiShakti

National Logistics Policy

ULIP-ready



DFD — LEVEL 1 • DATA → PREDICTION → OPTIMIZATION → DISPATCH



Closure / stale ETA → invalidate → candidates → re-solve → push new route.

INTEGRATION WITH EXISTING MUNICIPAL ICCC

Existing Municipal ICCC

NexFlow Adapter Layer

Freight Prediction

Curb / Route Optimization

Fleet + Driver Dispatch

ULIP PAYLOAD {sourceSystem:"NEXFLOW",
eventType:"VEHICLE_UPDATE", vehicleNo:"WB01XX0001"}

Integration with existing ICCC infra; ULIP access is authorization-gated.

OPEN-SOURCE BASE STACK • PostgreSQL + PostGIS + Python + OSRM

FAILURE + RESILIENCE • deterministic response paths

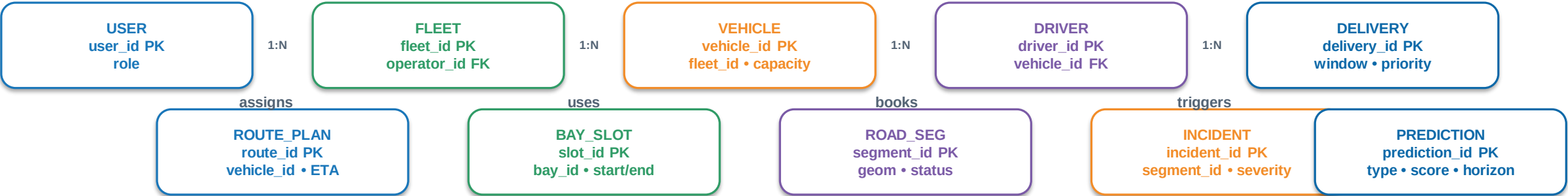
MISSED BAY WINDOW
Grace period → release slot → assign next feasible slot.

NETWORK FAILURE **OFFLINE DRIVER CACHE / SMS FALLBACK**
Store-and-forward GPS + cached route/slot; sync when online.

BAY UNEXPECTEDLY OCCUPIED
Conflict flag → nearest feasible bay/time → notify operator.

STALE / SPOOFED GPS Timestamp threshold → last-known position → human override.

ER DIAGRAM — CORE OPERATIONAL + GEO MODEL



Spatial store: PostGIS with GiST indexes • Hot state: Redis • Audit/event log: PostgreSQL • adapter-based integration keeps the core deployable.

TARGET HYPOTHESES • matched baseline vs NexFlow replay runs

-18%

Last-mile idle time

→ lower idle fuel consumption & emissions (expected secondary effect)

+25%

Loading-bay turnover

→ more completed deliveries per curb-hour

-10%

Distance / order

→ lower fuel + maintenance burden

-15%

Late orders

→ fewer penalties / failed delivery attempts

ECONOMIC IMPACT LOGIC

Idle minutes avoided ↓ Fuel avoided ↓ CO₂e avoided

Vehicle-km reduced ↓ Fuel + maintenance cost ↓

Bay turnover ↑ → more deliveries per curb-hour

BUSINESS MODEL — PREMIUM / VALUE-BASED ACCOUNTS

B2G • CITY ZONE PRO

₹5–15 L / year / zone

Municipal dashboard + forecasts + curb controls

B2B • FLEET PRO

₹150–300 / active vehicle / month

Dynamic routing + ETA protection + analytics

CURB ACCESS

₹10–20 / peak reserved slot

Transaction fee for booked loading access

GO-TO-MARKET • INTEGRATION WITH EXISTING MUNICIPAL ICCC INFRASTRUCTURE

PHASE 1 • PILOT

Burrabazar / Posta
1 zone • 100 vehicles
matched baseline replay

PHASE 2 • INTEGRATE

Municipal ICCC / TMC
fleet operators + curb management

PHASE 3 • EXPAND

Multi-zone → multi-city
enterprise fleet accounts

Deployment partners are potential: municipal ICCCs / Traffic Management Centres (e.g., KMC/KMDA, Delhi Traffic Police); no partnership is claimed.

GOVERNMENT GUIDELINES

- PM GatiShakti
- National Logistics Policy
- ULIP

DATA FEEDS

- Delhi GTFS / OTD
- Kochi Metro GTFS
- OpenStreetMap • Open-Meteo

ENGINE DOCUMENTATION

- OSRM
- OR-Tools
- PostGIS
- TomTom Traffic API

SCAN TO VERIFY PROJECT ACCESS



LIVE DEMO • GITHUB • API DOCS

DPIIT / PM GatiShakti + NLP

<https://www.dpiit.gov.in/offerings/schemes-and-services/details/pm-gati-shakti-kjNyATMtQWa>

Kochi GTFS

<https://corporate.kochimetro.org/open-data>

ULIP

<https://goulip.in/>

OSRM API

<https://project-osrm.org/docs/>

Delhi GTFS / OTD

<https://otd.delhi.gov.in/documentation/>

OR-Tools

<https://developers.google.com/optimization/routing>

ULIP direct access is authorization-gated. Adapter schema is a concrete technical artefact; no production-access claim. Pilot evidence: matched baseline vs NexFlow + road-closure replay.

Q&A FORMULATION | $\min \Sigma(\alpha \cdot \text{delay} + \beta \cdot \text{curb_wait} + \gamma \cdot \text{late} + \delta \cdot \text{distance} + \varepsilon \cdot \text{reroute})$ s.t. capacity, delivery windows, bay exclusivity, fleet limits, road-closure feasibility; binary $x(v,r)$, $y(v,b,t)$, $u(v,j)$.